# Ember Lu

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## Education

**University of California, Santa Cruz** — Santa Cruz, CA
*Bachelor of Science in Computer Science, 4.0 GPA* — *Sept. 2024 – Dec. 2027*

## Experience

**Software Engineering Intern** — June 2026 – Present
*Hewlett Packard Enterprise (HPE)* — *San Jose, CA*

- Engineering an autonomous multi-agent pipeline with LangGraph, GitHub Copilot, and a Model Context Protocol (MCP) integration, backed by a VectorDB-powered RAG knowledge base, to automate codebase context retrieval and generate bug-fix patches

**Software Engineering Intern** — Sept. 2025 – Dec. 2025
*Overture Maps Foundation* — *Santa Cruz, CA*

- Adapted 3+ lightweight language and embedding models (Meta Llama, Microsoft Phi-3, Google Electra) into sequence-classification architectures, matching 3,000+ geospatial data points with 90%+ accuracy
- Built PyTorch optimization pipelines with LoRA fine-tuning to improve model efficiency, while documenting OKR progress to maintain cross-team visibility into performance gains

**Software Engineering Intern** — Jul. 2025 – Sept. 2025
*PicnicHealth* — *San Francisco, CA*

- Shipped 2 foundational full-stack AI features (React, Node.js) that cut clinical trial record processing time by 95%
- Integrated 3+ API endpoints with REST controllers, real-time React/Apollo GraphQL components, Temporal workflow triggers, and PostgreSQL/Hasura data operations

## Projects

**Global Leadership Board for LM Geodata Performance** — July 2026 – Present

- Building one of the first global leaderboards benchmarking small, scalable language models against standardized validation datasets for geospatial conflation, expanding into additional real-world domains

**Syllabi Wrapped** | *Node.js, SQLite, 1st Place @ HackDay* — Feb. 2026

- Built a full-stack web app that uses GPT to parse uploaded course syllabi into structured data, backed by a Node.js server and SQLite database, with a user interface twist
- Designed a Spotify Wrapped-inspired swipe slideshow to present grade weighting, deadlines, and course policies; won 1st Place at CruzHacks Hack Day

**CalmSense** | *PyTorch, 3rd Place @ Santa Cruz AI Hackathon* — Jun. 2025

- Built a PyTorch neural network that analyzes real-time health data to predict oncoming anxiety attacks with 90% accuracy, with a web interface for guided breathing patterns to help alleviate symptoms

## Leadership

**President** | *Girls Who Code* — Jan. 2025 – Present
**Vice President** | *Google Developer Group* — Jun. 2025 – Present
**President** | *Santa Cruz Artificial Intelligence* — Aug. 2025 – Present
**Co-Lead** | *Hack.dev (NVIDIA x ASUS Hackathon)* — April 2026 – Present

## Technical Skills

**Languages**: Python, TypeScript, Java, C/C++, SQL, R, JavaScript, HTML/CSS, English, Mandarin
**Frameworks/Technologies**: React, Node.js, Flask, Hasura, PostgreSQL, Docker, Linux, PyTorch, LangGraph
**Developer Tools**: Git, Google Cloud Platform, Temporal, Linear, Figma, Microsoft Office, GitHub Copilot
**Relevant Coursework**: Data Structures and Algorithms, Machine Learning, Computer Systems Design, Natural Language Processing, Database Systems, Computer Systems & C Programming, Mathematical Thinking for Computer Science, Probability and Statistics, Analysis of Algorithms